Supplementary Material S4
Complete Stratified Synthesis of the Corpus

Facial Emotion Recognition through Machine Learning in Children with Autism Spectrum Disorder: A systematic scoping review of open-access computational evidence and of the gaps toward therapeutic monitoring

This document complements Table 17 in the manuscript (11 studies of greatest translational traceability, Section 6.3). Section A presents the 21 eligible studies organised by clinical domain (D1–D4); Section B presents the 9 contextual studies outside the eligible set; Section C presents the systematic longitudinal tracking audit and the multi-label coding matrix for all 21 eligible studies.

The internal # column corresponds to the corpus order shared with Supplementary Material S2 (C prefix for contextual evidence), not to bibliographic reference numbers. Gaps in the sequence are not numbering errors: numbers were assigned before the final integrity and eligibility verification and are not reused. Ahmad et al. (2024), retracted after corpus finalisation (17 August 2026, *Healthc Technol Lett* 2026;13(1):e70095), is excluded with retroactive effect. The 4 retracted studies do not appear in this document.

Section A. Eligible corpus (21 studies)

#	Study	Clinical task	Dataset	Population / Setting	Architecture	Metrics	Data prov.
Domain D1: Automated facial expression (7 studies)							
1	Johnson et al. (2025)	Emotion recogn.	AffectNet + clinical	ASD children (N=48) / Lab./Ecol.	EfficientNet-v2	ACC 67.8%[l]	Clinical
8	Radočaj & Mart. (2025)	Emotion recogn.	FERAC	ASD children / Lab.	Swin Transformer	ACC 80%, F1 78.9%	Partial prov.
9	Washington et al. (2022)	Emotion recogn.	CAFE Sub. A + clinical	Children / Lab./Ecol.	ResNet-152	Bal.ACC 79.1%, F1 78%	Clinical
10	Sholikah et al. (2024)	Real-time recogn.	FER-2013	N=3 ASD (test)[a] / Mobile	VGG-16	ACC 91%, F1 91%	Public gen.
21	Aljbori et al. (2025)	Emotion recogn.	Kaggle ASD	ASD children / Lab.	3D-CNN + LSTM	ACC 99.01%[j], F1 98.6%	Web
27	Dapogny et al. (2019)	Facial expr.	JEMImE (N=157)	Children 6–11 / Clin. game	Rand. Forests + PCRF	ACC 81.9%	Public cur.
28	Leo et al. (2018)	Facial prod. ASD	CK+ + clinical (N=17)	ASD children / Clinical	HOG + SVM + CNN	TP 85% (CK+)[b]	Clinical
Domain D2: Affect and therapeutic engagement (4 studies)							
4	Lecciso et al. (2021)	Therapeutic interv.	Private (N=24)	ASD children / Therapy	OpenFace	g=0.52	Clinical
15	Rudovic et al. (2018)	Affect / therapy	MIT private (N=35)	ASD children / Robot th.	PPA-net	ICC ∼60%	Clinical
16	Cicceri et al. (2026)	Pointing gestures	Private clinical	ASC children / Clinical	4-level framework	mAP 88.6%	Clinical
17	Rudovic et al. (2019)	Active learning	MIT private (N=35)	ASD children / Robot th.	4 LSTMs + Deep RL	ACC 82.3%, F1 52.6%[n]	Clinical
Domain D3: Emotion recognition intervention (3 studies)							
2	He et al. (2019)	Eye-tracking	Private (N=42)	ASD preschoolers n=21 / Clinical	Eye-tracking	r=0.619	Clinical
3	Reed et al. (2021)	Epidemiological	ALSPAC (N=3,562)	Paediatric cohort / Population	Regression	b=−0.18	Cohort
18	Almeida et al. (2019)	Serious game	No ML dataset	ASD 6–12 / Game	RPG (no ML)	ISO/IEC usab.	N/A
Domain D4: Screening or diagnosis of ASD (7 studies)							
5	Shaik et al. (2025)	ASD detection	Kaggle ASD	Children / Lab.	Ensemble CNN	ACC 84%, F1 84.2%	Web
12	Sariyanidi et al. (2025)	Autism pred.	CHOP clinical (N=42)	ASD + TD / Clinical	Dict. Learn. + SVM	Fig.[c]	Clinical
13	Hassan et al. (2025)	ASD level assess.	DREAM (N=61)	ASD children 3–6 / Robot th.	RF + XGBoost + ExtraT.	ACC 95.6%, F1 92.8%[o]	Public cur.
19	Alam et al. (2025)	ASD detection	Kaggle + YTUIA; UIFID test	Children / Lab.	ASD-UANet	ACC 96%	Web
20	Hwang et al. (2024)	DD screening	Private (N=89; 33 DD, 56 TD)	34–77 months / Mobile	LSTM + 2D-FAN	ACC 88%, F1 80%	Clinical
23	Vedantham et al. (2025)	ASD detection	Kaggle	Children / Lab.	GWO-GEfficientNet	ACC 96.5%, F1 96.4%	Web
24	Hosney et al. (2024)	ASD detection	Kaggle (co-auth.)[d]	Children / Lab.	YOLOv8 + DCNN	ACC 97.2%, F1 93%	Web

Section B. Contextual evidence outside the eligible set (9 studies)

Five infringe CE(3) (exclusively non-facial data), three infringe the age criterion (exclusively adult participants) and one infringes CI(3) (journal not indexed in SJR in the year of publication).

#	Study	Clinical task	Dataset	Population	Setting	Architecture	Metrics	Data prov.
Adult participants (3 studies)								
C1	El Rhatassi et al. (2024)	Emotion recogn.	FER-2013 + Kaggle autism	Autistic adults	Lab.	CNN vs ViT	ACC 65–74%	Public gen.
C2	El Rhatassi et al. (2025)	Real-time recogn.	CK+ + HeyGen AI	Autistic adults	Lab.	CNN×3 + LSTM + CC	ACC val. 53%	Mixed
C3	Levy et al. (2024)	Wearable system	FER-2013	ASD adults (N=10)	Ecol.	CNN + HW wearable	Usability	Public gen.
Non-facial data only, CE(3) (5 studies)								
C4	Nagae & Lee (2022)	Therapy + EDA	Private EDA (N=11)	DD children	Robot th.	EDA + wavelet + SVM	Conf. matrix	Clinical
C5	Papaioannou et al. (2021)	EEG cogn. load	EEG (N=63)	ASD + ADHD adults	Clinical	MSE + PLSC	ANOVA	Clinical
C6	Xu & Ju (2023)	VBM brain	ABIDE (N=203)	ASD 8–12 (n=98)	Clinical	VBM	GMV+ADOS corr.	Public cur.
C7	O'Sullivan et al. (2023)	Speaker diariz.	Clin. trial audio (N=72)	ASD 5–45	Home	Pyannote + xvector	F1 76.5%	Clinical
C8	Alsubai (2025)	ASD detect. (tab.)	Kaggle screening, text, clinical	12–36 m; adults 60–96	Screening	Transformer MDNCT	ACC 94.7%	Web
Journal not indexed in SJR, CI(3) (1 study)								
C9	Akın-Bülbül et al. (2024)	ET screening	Private ET	Infants 18–36 m	Clinical	KNN + features	ACC 81.45%	Clinical

Section C. Longitudinal tracking audit and multi-label coding matrix

Section C.1 presents the systematic longitudinal tracking audit for all 21 eligible studies, evaluating the presence of intraindividual temporal series, progress outcomes, test-retest reliability, and responsiveness to change. Section C.2 presents the multi-label coding matrix, documenting the objective variable type, output class structure, and training representation for each study.

The principal finding confirmed by the audit is that no study in the eligible corpus reports session-to-session longitudinal tracking variables. Studies with multiple recorded sessions do not fill this gap: Rudovic et al. (2018, 2019) (35 children, repeated robot interactions) measure affect or engagement per session independently, and Hassan et al. (2025) (61 children, 3,121 sessions, DREAM dataset) use the session data to predict the ASD level and the ADOS score of each child; none quantifies intraindividual change trajectories, test-retest reliability, or responsiveness. Lecciso et al. (2021) is the only study with a pre-post design; it compares two time points but does not report intraindividual session-by-session series. Longitudinal therapeutic monitoring therefore remains an open gap.

C.1. Longitudinal tracking audit (21 eligible studies)

Columns: *Sessions*: number of distinct measurement points or sessions in the study design; *Subject unit*: unit of analysis (individual vs group); *Intraindiv. series*: intraindividual temporal series across sessions (Yes / No); *Progress outcomes*: session-to-session progress indicators (Yes / No); *Test-retest*: test-retest reliability coefficient reported (Yes / No); *Responsiveness*: sensitivity to change across time or intervention (Yes / No).

#	Study	Sessions	Subject unit	Intraindiv. series	Progress outcomes	Test-retest	Responsiveness
Domain D1 (7 studies)							
1	Johnson et al. (2025)	1 (cross-sect.)	Group	No	No	No	No
8	Radočaj & Mart. (2025)	1 (cross-sect.)	Group	No	No	No	No
9	Washington et al. (2022)	1 (cross-sect.)	Group	No	No	No	No
10	Sholikah et al. (2024)	1 (usability test)	Individual[e]	No	No	No	No
21	Aljbori et al. (2025)	1 (cross-sect.)	Group	No	No	No	No
27	Dapogny et al. (2019)	1 (clinical game)	Group	No	No	No	No
28	Leo et al. (2018)	1 (cross-sect.)	Group (N=17)	No	No	No	No
Domain D2 (4 studies)							
4	Lecciso et al. (2021)	2 (pre-post)	Individual	Pre-post only[f]	No	ND	ND
15	Rudovic et al. (2018)	Multiple[h]	Individual	No[h]	ND	ND	ND
16	Cicceri et al. (2026)	1 (clinical)	Group	No	No	No	No
17	Rudovic et al. (2019)	Multiple[h]	Individual	No[h]	ND	ND	ND
Domain D3 (3 studies)							
2	He et al. (2019)	1 (cross-sect.)	Individual	No	No	No	No
3	Reed et al. (2021)	Cohort (2 timepoints)	Population	No[i]	ND	ND	ND
18	Almeida et al. (2019)	1 (game session)	Group	No	No	No	No
Domain D4 (7 studies)							
5	Shaik et al. (2025)	1 (cross-sect.)	Group	No	No	No	No
12	Sariyanidi et al. (2025)	1 (screening)	Group (N=42)	No	No	No	No
13	Hassan et al. (2025)	3,121 sessions total	Group (N=61)	No[g]	ND	ND	ND
19	Alam et al. (2025)	1 (cross-sect.)	Group	No	No	No	No
20	Hwang et al. (2024)	1 (screening)	Group (N=89)	No	No	No	No
23	Vedantham et al. (2025)	1 (cross-sect.)	Group	No	No	No	No
24	Hosney et al. (2024)	1 (cross-sect.)	Group	No	No	No	No

C.2. Multi-label coding matrix (21 eligible studies)

Columns: *Obj. variable*: type of target variable; *Output classes*: number of output categories; *Training repr.*: primary training representation modality; *Stratified val.*: stratified or subject-independent validation reported; *Multi-label*: whether the study reports multi-label (simultaneous assignment of ≥2 non-exclusive labels) as opposed to multi-class or binary single-label classification.

#	Study	Obj. variable	Output classes	Training repr.	Stratified val.	Multi-label
Domain D1 (7 studies)						
1	Johnson et al. (2025)	Multi-class emotion	7 (basic emotions)	CNN / AffectNet+clinical	No	No
8	Radočaj & Mart. (2025)	Multi-class emotion	4 (natural, fear, joy, anger)	Transformer / FERAC	Yes[m]	No
9	Washington et al. (2022)	Multi-class emotion	7 (basic emotions)	CNN / CAFE subset	Yes	No

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#	Study	Obj. variable	Output classes	Training repr.	Stratified val.	Multi-label
10	Sholikah et al. (2024)	Multi-class emotion	7 (basic emotions)	CNN / FER-2013	No	No
21	Aljbori et al. (2025)	Multi-class emotion	6 (basic emotions)	3D-CNN + LSTM / Kaggle	No	No
27	Dapogny et al. (2019)	Multi-class emotion	4 (game expressions)	Rand. Forests / JEMImE	Yes	No
28	Leo et al. (2018)	Multi-class emotion	7 + ASD prod.	HOG+CNN / CK+ + clinical	N/A	No
Domain D2 (4 studies)						
4	Lecciso et al. (2021)	Continuous affect	N/A (AU intensity)	OpenFace / clinical	N/A	No
15	Rudovic et al. (2018)	Continuous affect	N/A (ICC scale)	PPA-net / MIT clinical	No[p]	No
16	Cicceri et al. (2026)	Multi-class gesture	4 (pointing stages)	Spatial framework / clinical	No	No
17	Rudovic et al. (2019)	Multi-class engagement	3 (low/med/high)	LSTM+RL / MIT clinical	Yes	No
Domain D3 (3 studies)						
2	He et al. (2019)	Correlation (continuous)	N/A	Eye-tracking	N/A	No
3	Reed et al. (2021)	Continuous (regression)	N/A	Population regression	N/A	No
18	Almeida et al. (2019)	Usability (non-ML)	N/A	No ML classifier	N/A	No
Domain D4 (7 studies)						
5	Shaik et al. (2025)	Binary (ASD / TD)	2	CNN ensemble / Kaggle	Yes[p]	No
12	Sariyanidi et al. (2025)	Binary (ASD / TD)	2 (graphical)	Dict. learn. / CHOP	Yes[p]	No
13	Hassan et al. (2025)	ASD level (multi-class)[g]	ND[g]	RF+XGBoost / DREAM	ND[g]	No
19	Alam et al. (2025)	Binary (ASD / TD)	2	CNN ensemble / Kaggle+YTUIA	Yes	No
20	Hwang et al. (2024)	Binary (DD / TD)	2	LSTM+2D-FAN / clinical	Yes	No
23	Vedantham et al. (2025)	Binary (ASD / TD)[k]	2	CNN hybrid / Kaggle	No	No
24	Hosney et al. (2024)	Binary (ASD / TD)	2	YOLO+CNN / Kaggle	No	No

Notes.

a. 3 ASD participants in Sholikah et al. (2024) involved in a usability test, not in metric computation.

b. TP 85% for Leo et al. (2018) corresponds to the CK+ public dataset of typically developing adults, not to the 17 ASD children.

c. Fig.: metrics reported in graphical format only, without numerically extractable values. Applies to Sariyanidi et al. (2025).

d. (co-auth.): Kaggle dataset created by a co-author of Hosney et al. (2024), constituting a documented conflict of interest.

e. N=3 ASD children; usability test only (see note a).

f. Lecciso et al. (2021) employs a pre-post design comparing two time points (g=0.52) but does not report intraindividual session-by-session series, test-retest reliability, or responsiveness.

g. Hassan et al. (2025) use the 3,121 robot-enhanced therapy sessions of the 61 children in the DREAM dataset to predict the ASD level (classification) and the ADOS score (regression, RMSE 1.78, R^2 0.74). Because the outcome is the severity of the condition and not engagement or affect during the session, the study is assigned to domain D4. Indexed excerpts of the article mention three ADOS-based categories (non-autism, autism spectrum disorder and autism), but the full text could not be accessed to confirm the number of levels used for classification, the cut-off points or the validation scheme; these entries are therefore ND. The study does not quantify intraindividual change trajectories.

h. Rudovic et al. (2018, 2019) involve repeated interactions with a social robot (N=35); the design does not quantify intraindividual change trajectories, test-retest reliability, or responsiveness to intervention.

i. Reed et al. (2021) uses the ALSPAC epidemiological cohort (N=3,562); regression captures cross-sectional associations; no intraindividual session series are reported.

j. Aljbori et al. (2025) report 99.01% accuracy in their results table and 99.82% in the abstract; the table value is retained here.

k. Vedantham et al. (2025) classify autistic versus non-autistic facial images; an auxiliary facial expression dataset is relabelled into three classes (positive, negative, neutral) without a stated mapping to the autism labels.

l. The 67.8% reported by Johnson et al. (2025) corresponds to seven-class classification on AffectNet.

m. Radočaj and Martinović (2025) state an 80/20 stratified train-validation split, while the image counts they report are inconsistent (691/79 in one section and 615/155 in another).

n. Best values after adaptation in Rudovic et al. (2019): multimodal Q-learning with model-level fusion, three engagement levels (Tables 2 and 3 of the article), with training and test children kept separate.

o. The 95.6% accuracy of Hassan et al. (2025) corresponds to the boosted 3D biomarker approach for autism-level prediction; adding saliency maps, the full framework reports 97.36% accuracy and an F1-score of 95.56%.

p. Validation schemes verified in the full texts: Shaik et al. (2025) use stratified 5-fold cross-validation together with a stratified 80/20 split; Sariyanidi et al. (2025) use leave-one-out cross-validation by participant with a nested 5-fold inner loop for the SVM parameter; Rudovic et al. (2018) split each child's own data 40/20/40 into training, validation and test sets (personalised models), so validation is neither stratified nor subject-independent.

1. The internal # column uses the corpus order shared with Supplementary Material S2 for eligible studies and the C prefix for contextual evidence. It does not correspond to bibliographic reference numbers, and gaps in the sequence are not numbering errors.
2. Retracted studies (Güngör et al., 2025; Mahmood et al., 2025; Kumar et al., 2024; Ahmad et al., 2024) do not appear. Ahmad et al. (2024) was retracted on 17 August 2026 (*Healthc Technol Lett* 2026;13(1):e70095); all four are documented in Section 2.7 of the manuscript.
3. Bal.ACC for Washington et al. (2022): balanced accuracy 79.1% on CAFE subset with human agreement $\geq$60%; full CAFE set value is 66.9%.
4. Almeida et al. (2019): systems design without ML classification; ISO/IEC usability metrics are not comparable with supervised classification metrics.
5. Contextual studies in Section B were appraised with MMAT in Supplementary Material S2 (Section B) but do not participate in the aggregate MMAT distribution of the manuscript.